

doses of psychotropic drugs previously prescribed. At the end of α -tocopherol therapy the severity of TD was decreased by 12-75% (mean 49%) according to the AIMS scale.

A 3-fold increase (compared with healthy donors) in the plasma content of malonic dialdehyde (MDA), an end product of lipid peroxidation (LPO), was recorded in patients prior to α -tocopherol administration. The blood contents of α -tocopherol and ascorbate did not differ from those in healthy donors, whereas superoxide dismutase (SOD) activity in erythrocytes was lower than normal. At the end of α -tocopherol therapy

the blood MDA content was lowered and the activity of the antioxidant system was increased: the blood content of ascorbate increased by 31% ($p < 0.1$) and that of α -tocopherol increased by 52% ($p < 0.01$). The activity of SOD also increased, reaching the control values. A correlation was established between the decrease in MDA concentration and the clinical effect produced by α -tocopherol therapy ($r = 0.87$, $p < 0.01$).

Taken together, our findings indicate that there is a noncompensatory intensification of LPO in TD patients and that α -tocopherol therapy evidently normalizes LPO in these patients.

Anxiety, Atherogenesis, and Antioxidant Protection: Clinico-Pathogenetic Relationships

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In the course of studying and comparing involutionary-climacteric psychoses by a number of clinico-biological parameters, we revealed opposite tendencies in the intensity of endocrine-metabolic processes, biological oxidation, and lipid metabolism disorders, as well as differences related to sex, intellectual or physical work, residential environment (urban or rural), etc. Only one common trend was noted - the age of manifestation (45-60 years), although the incidence of atherosclerosis is higher in middle-aged individuals and declines again in old age [G. G. Avtandilov, 1972; B. A. Asetto, 1972; K. Z. Tnimova, 1972; A. M. Vikhert and V. S. Zhdanov, 1976; A. M. Lifshits *et al.*, 1988]. In contrast to atherosclerotic pathologies, climacteric psychoses are reversible, despite a sustained anxiety that models chronic stress. Emotional tension intensifies lipid peroxidation [M. V. Poyuroskii *et al.*, 1986; G. G. Neznamov *et al.*, 1986; Yu. A. Aleksandrovskii *et al.*, 1988] and stimulates

the antioxidant systems of nonspecific resistance of the organism to stress factors [V. V. Sokolovskaya *et al.*, 1987; V. N. Bobyrev, V. F. Pochernyaeva, 1991]. Analyzing our clinical observations and the role of minor lipid components (naphthoquinones and tocopherols) in the regulation of the phospholipid/cholesterol ratio in biological membranes, we put forward a hypothesis that atherogenesis generates not only a damaging but also a protective potential [S. M. Plotnikov and V. A. Lider, 1992], the biological significance of which consists in antioxidant membrane stabilizing functions. We suggested that atherogenesis "kicks in" when other endogenous possibilities are exhausted or premorbidly insufficient for gentle dampening of the "metabolic impact" during the climacteric period. There are sufficient grounds for using of antioxidants to prevent and treat involutionary and vascular psychic disorders, even though these pathologies are fundamentally different.

A New Approach to the Treatment of Endogenous Depressions with Antioxidants

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In order to develop new approaches to the optimization of the therapy of endogenous depressions, we tested the possibility of using antioxidants (α -tocopherol and emoxipine) in combination with antidepressants. The study was carried out by the open method in middle-aged and elderly patients resistant to anti-

depressive therapy. In middle-aged patients emoxipine and α -tocopherol (600 mg/day) considerably reduced the manifestations of depressive symptoms assessed by the HAMD scale (in 53.3 and 66.7% of the patients, respectively) and abolished the resistance to tri- and tetracyclic antidepressants. In elderly patients, α -toco-